

Artificial Intelligence Programming Languages

Student Notes

1. Introduction

- ▶ AI languages are used to build intelligent systems.
 - ▶ They support reasoning, learning, and data processing.
 - ▶ Different AI tasks (like machine learning, natural language processing, expert systems, robotics, etc.) benefit from different languages because each has strengths in areas like speed, flexibility, symbolic reasoning, or mathematical computation.

2. Why AI Needs Special Languages

- ▶ Symbolic reasoning (working with facts, rules & logic)
- ▶ Handling uncertainty
- ▶ Learning from data
- ▶ Pattern recognition
- ▶ Massive mathematical computations
- ▶ Manipulating knowledge representations

Because of this, AI languages must support:

- ▶ Fast mathematical operations
- ▶ Efficient memory handling
- ▶ Easy data structure manipulation
- ▶ Logical rule-based coding
- ▶ Machine learning libraries
- ▶ Flexibility for experimentation

Major AI Programming Languages

- ▶ **A. LISP (List Processing Language)**
- ▶ **One of the earliest AI languages (1958).**
Created by John McCarthy.
- ▶ **Key Features**
- ▶ Excellent for symbolic reasoning
- ▶ Code is written in lists → very flexible
- ▶ Automatic memory management
- ▶ Supports recursion well
- ▶ **Used For**
- ▶ Expert systems
- ▶ Natural Language Processing
- ▶ Research and prototyping
- ▶ **Why it's important**
- ▶ LISP introduced many ideas used in modern languages (dynamic typing, garbage collection).

PROLOG (Programming in Logic)

- ▶ Developed in 1970s.
- ▶ **Key Features**
- ▶ Based on **logic** and **rules**
- ▶ You write *facts and rules*, not step-by-step instructions
- ▶ PROLOG “searches” for answers automatically
- ▶ **Used For**
- ▶ Expert systems
- ▶ Language processing
- ▶ Solving logical puzzles
- ▶ Knowledge-based reasoning
- ▶ **Why it's unique**
- ▶ You describe *what* you want, not *how* to do it.

Python

- ▶ **Key Features**

- ▶ Easy to read and write
- ▶ Huge number of libraries for AI
- ▶ Works across all operating systems
- ▶ Great community support

- ▶ **Common Libraries**

- ▶ NumPy → mathematics
- ▶ TensorFlow / PyTorch → machine learning
- ▶ Pandas → data processing
- ▶ Scikit-learn → ML models
- ▶ OpenCV → computer vision
- ▶ NLTK / spaCy → natural language processing

- ▶ **Used For**

- ▶ Machine learning
- ▶ Deep learning
- ▶ Data science
- ▶ Robotics
- ▶ Chatbots

Java

- ▶ **Key Features**
- ▶ Fast, secure, cross-platform
- ▶ Great for large enterprise systems
- ▶ Strong object-oriented design
- ▶ **Used For**
- ▶ AI in banking & enterprise applications
- ▶ Search engines
- ▶ Web-based AI tools
- ▶ Multi-threaded applications
- ▶ **Why used in AI**
- ▶ Reliable performance and structure.

C

- ▶ **Key Features**
- ▶ Very fast execution
- ▶ Good for low-level operations
- ▶ Great memory control
- ▶ **Used For**
- ▶ Robotics
- ▶ Embedded AI
- ▶ Real-time systems (e.g., self-driving cars)
- ▶ Game AI
- ▶ **Why it matters**
- ▶ Speed is important for time-sensitive AI tasks.

Choosing a language

Task

Machine Learning

Deep Learning

Robotics

Web AI

Logic-Based AI / Expert Systems

High-Performance Systems

Enterprise AI

Best Languages

Python, R

Python

C++, Python

JavaScript

PROLOG, LISP

C++

Java

Key Characteristics of Good AI Languages

- ▶ A strong AI language should offer:
- ▶ **Easy handling of complex data structures**
- ▶ **Powerful math libraries**
- ▶ **Strong community support**
- ▶ **Fast execution (for real-time tasks)**
- ▶ **Flexibility for experimentation**
- ▶ **Integration with AI frameworks**
- ▶ No single language is perfect.
Developers choose based on the task.

10. Summary

- ▶ Different AI tasks need different languages.
 - ▶ Modern AI dominated by Python.
 - ▶ Logic-based AI uses PROLOG and LISP.